

ABIDS Hypothesis & Statistical Results

Every Tested Hypothesis, Explained Plainly

Aaryatech Behavioral Joining Intelligence -- ABIDS Demonstration Run

This analysis uses a purpose-built DEMONSTRATION DATASET created to prove the ABIDS method end to end. It is not V Dart's or any client's real historical data.

14 hypotheses were proposed; 11 had enough supporting evidence to test. Every test below used only the discovery and hypothesis_generation splits -- held_out_test was never opened.

1. Status Quo Bias (BM-001)

Question

Is there a difference in joining outcome between applications with vs. without evidence of Status Quo Bias?

What We Compared

Group sizes: with_mechanism=199;without_mechanism=1475. Sample size: 1674.

What Test We Used

chi_square_test, chosen automatically based on group sizes and expected cell counts.

What The Result Means

p-value = 0.0198 (Benjamini-Hochberg adjusted: 0.0437). Direction: lower joining with mechanism.

How Strong The Difference Is

Effect size: 0.0569 (Cramer's V). Evidence grade: WEAK.

What We Can Say

This association is statistically significant but the effect size is small -- worth monitoring, not yet worth major action.

What We Cannot Say

Outcome operationalized as binary joined-vs-not-joined (simplified from the 6-category final_disposition). Association only -- not a causal claim. held_out_test was not used. This is an association only; it does not prove that Status Quo Bias causes candidates to join or not join.

2. Loss Aversion (BM-002)

Question

Is there a difference in joining outcome between applications with vs. without evidence of Loss Aversion?

What We Compared

Group sizes: with_mechanism=105;without_mechanism=1569. Sample size: 1674.

What Test We Used

chi_square_test, chosen automatically based on group sizes and expected cell counts.

What The Result Means

p-value = 0.0000 (Benjamini-Hochberg adjusted: 0.0000). Direction: lower joining with mechanism.

How Strong The Difference Is

Effect size: 0.1978 (Cramer's V). Evidence grade: SUPPORTED.

What We Can Say

This association is both statistically significant and has a meaningful effect size -- worth taking seriously as a candidate for further investigation.

What We Cannot Say

Outcome operationalized as binary joined-vs-not-joined (simplified from the 6-category final_disposition). Association only -- not a causal claim. held_out_test was not used. This is an association only; it does not prove that Loss Aversion causes candidates to join or not join.

3. Choice Overload (BM-005)

Question

Is there a difference in joining outcome between applications with vs. without evidence of Choice Overload?

What We Compared

Group sizes: with_mechanism=54;without_mechanism=1620. Sample size: 1674.

What Test We Used

chi_square_test, chosen automatically based on group sizes and expected cell counts.

What The Result Means

p-value = 0.0472 (Benjamini-Hochberg adjusted: 0.0866). Direction: lower joining with mechanism.

How Strong The Difference Is

Effect size: 0.0485 (Cramer's V). Evidence grade: INCONCLUSIVE.

What We Can Say

This result did not reach both significance and a meaningful effect size in this pass.

What We Cannot Say

Outcome operationalized as binary joined-vs-not-joined (simplified from the 6-category final_disposition). Association only -- not a causal claim. held_out_test was not used. This is an association only; it does not prove that Choice Overload causes candidates to join or not join.

4. Friction (BM-006)

Question

Is there a difference in joining outcome between applications with vs. without evidence of Friction?

What We Compared

Group sizes: with_mechanism=299;without_mechanism=1375. Sample size: 1674.

What Test We Used

chi_square_test, chosen automatically based on group sizes and expected cell counts.

What The Result Means

p-value = 0.0002 (Benjamini-Hochberg adjusted: 0.0005). Direction: higher joining with mechanism.

How Strong The Difference Is

Effect size: 0.0915 (Cramer's V). Evidence grade: WEAK.

What We Can Say

This association is statistically significant but the effect size is small -- worth monitoring, not yet worth major action.

What We Cannot Say

Outcome operationalized as binary joined-vs-not-joined (simplified from the 6-category final_disposition). Association only -- not a causal claim. held_out_test was not used. This is an association only; it does not prove that Friction causes candidates to join or not join.

5. Reciprocity (BM-011)

Question

Is there a difference in joining outcome between applications with vs. without evidence of Reciprocity?

What We Compared

Group sizes: with_mechanism=148;without_mechanism=1526. Sample size: 1674.

What Test We Used

chi_square_test, chosen automatically based on group sizes and expected cell counts.

What The Result Means

p-value = 0.9006 (Benjamini-Hochberg adjusted: 0.9006). Direction: higher joining with mechanism.

How Strong The Difference Is

Effect size: 0.0031 (Cramer's V). Evidence grade: INCONCLUSIVE.

What We Can Say

This result did not reach both significance and a meaningful effect size in this pass.

What We Cannot Say

Outcome operationalized as binary joined-vs-not-joined (simplified from the 6-category final_disposition). Association only -- not a causal claim. held_out_test was not used. This is an association only; it does not prove that Reciprocity causes candidates to join or not join.

6. Anchoring (BM-013)

Question

Is there a difference in joining outcome between applications with vs. without evidence of Anchoring?

What We Compared

Group sizes: with_mechanism=8;without_mechanism=1666. Sample size: 1674.

What Test We Used

fishers_exact_test, chosen automatically based on group sizes and expected cell counts.

What The Result Means

p-value = 0.4591 (Benjamini-Hochberg adjusted: 0.5611). Direction: lower joining with mechanism.

How Strong The Difference Is

Effect size: 0.5270 (odds ratio). Evidence grade: INCONCLUSIVE.

What We Can Say

This result did not reach both significance and a meaningful effect size in this pass.

What We Cannot Say

Outcome operationalized as binary joined-vs-not-joined (simplified from the 6-category final_disposition). Association only -- not a causal claim. held_out_test was not used. This is an association only; it does not prove that Anchoring causes candidates to join or not join.

7. Confirmation Bias (BM-014)

Question

Is there a difference in joining outcome between applications with vs. without evidence of Confirmation Bias?

What We Compared

Group sizes: with_mechanism=61;without_mechanism=1613. Sample size: 1674.

What Test We Used

chi_square_test, chosen automatically based on group sizes and expected cell counts.

What The Result Means

p-value = 0.6611 (Benjamini-Hochberg adjusted: 0.7272). Direction: higher joining with mechanism.

How Strong The Difference Is

Effect size: 0.0107 (Cramer's V). Evidence grade: INCONCLUSIVE.

What We Can Say

This result did not reach both significance and a meaningful effect size in this pass.

What We Cannot Say

Outcome operationalized as binary joined-vs-not-joined (simplified from the 6-category final_disposition). Association only -- not a causal claim. held_out_test was not used. This is an association only; it does not prove that Confirmation Bias causes candidates to join or not join.

8. Identity Signaling (BM-018)

Question

Is there a difference in joining outcome between applications with vs. without evidence of Identity Signaling?

What We Compared

Group sizes: with_mechanism=84;without_mechanism=1590. Sample size: 1674.

What Test We Used

chi_square_test, chosen automatically based on group sizes and expected cell counts.

What The Result Means

p-value = 0.0000 (Benjamini-Hochberg adjusted: 0.0000). Direction: higher joining with mechanism.

How Strong The Difference Is

Effect size: 0.1182 (Cramer's V). Evidence grade: SUPPORTED.

What We Can Say

This association is both statistically significant and has a meaningful effect size -- worth taking seriously as a candidate for further investigation.

What We Cannot Say

Outcome operationalized as binary joined-vs-not-joined (simplified from the 6-category final_disposition). Association only -- not a causal claim. held_out_test was not used. This is an association only; it does not prove that Identity Signaling causes candidates to join or not join.

9. Commitment / Consistency (BM-020)

Question

Is there a difference in joining outcome between applications with vs. without evidence of Commitment / Consistency?

What We Compared

Group sizes: with_mechanism=138;without_mechanism=1536. Sample size: 1674.

What Test We Used

chi_square_test, chosen automatically based on group sizes and expected cell counts.

What The Result Means

p-value = 0.0000 (Benjamini-Hochberg adjusted: 0.0000). Direction: higher joining with mechanism.

How Strong The Difference Is

Effect size: 0.1289 (Cramer's V). Evidence grade: SUPPORTED.

What We Can Say

This association is both statistically significant and has a meaningful effect size -- worth taking seriously as a candidate for further investigation.

What We Cannot Say

Outcome operationalized as binary joined-vs-not-joined (simplified from the 6-category final_disposition). Association only -- not a causal claim. held_out_test was not used. This is an association only; it does not prove that Commitment / Consistency causes candidates to join or not join.

10. Goal Gradient (BM-023)

Question

Is there a difference in joining outcome between applications with vs. without evidence of Goal Gradient?

What We Compared

Group sizes: with_mechanism=174;without_mechanism=1500. Sample size: 1674.

What Test We Used

chi_square_test, chosen automatically based on group sizes and expected cell counts.

What The Result Means

p-value = 0.2605 (Benjamini-Hochberg adjusted: 0.3582). Direction: higher joining with mechanism.

How Strong The Difference Is

Effect size: 0.0275 (Cramer's V). Evidence grade: INCONCLUSIVE.

What We Can Say

This result did not reach both significance and a meaningful effect size in this pass.

What We Cannot Say

Outcome operationalized as binary joined-vs-not-joined (simplified from the 6-category final_disposition). Association only -- not a causal claim. held_out_test was not used. This is an association only; it does not prove that Goal Gradient causes candidates to join or not join.

11. Uncertainty / Ambiguity Aversion (BM-029)

Question

Is there a difference in joining outcome between applications with vs. without evidence of Uncertainty / Ambiguity Aversion?

What We Compared

Group sizes: with_mechanism=337;without_mechanism=1337. Sample size: 1674.

What Test We Used

chi_square_test, chosen automatically based on group sizes and expected cell counts.

What The Result Means

p-value = 0.2520 (Benjamini-Hochberg adjusted: 0.3582). Direction: lower joining with mechanism.

How Strong The Difference Is

Effect size: 0.0280 (Cramer's V). Evidence grade: INCONCLUSIVE.

What We Can Say

This result did not reach both significance and a meaningful effect size in this pass.

What We Cannot Say

Outcome operationalized as binary joined-vs-not-joined (simplified from the 6-category final_disposition). Association only -- not a causal claim. held_out_test was not used. This is an association only; it does not prove that Uncertainty / Ambiguity Aversion causes candidates to join or not join.